DA3410: Assignment 11

1 Objectives

- Use the six simulated data sets from Section 5 of the paper: Linear, Friedman 1, Friedman 2, Friedman 3, Rotated Sine, and Soft Radial. Each data set has 5 predictors drawn uniformly from $[0, 1]$. The response is a known function of the predictors plus Gaussian noise. Use training sizes of 500, 1000, 2500, and 5000, signal-to-noise ratios of 0.5, 1, 2, and 3, and a test set of 1000 points.
- Design controlled experiments and change one factor at a time.
- Measure the results, plot them, and compare them with the results in the paper.
- Explain the results in clear and simple terms. State where your results agree with the paper and where they differ.

2 Ordinary Least Squares as an Attention Mechanism

Summary of Paper

This paper shows that the predictions of ordinary least squares (OLS) can be written as the output of a simple attention module. Write $(X'X)^{-1} = U\Lambda^{-1}U'$ and encode the predictors as $F = XU\Lambda^{-1/2}$. The OLS prediction is then a weighted sum of the training outcomes, $\hat{y}_{test} = [F_{test}F'_{train}]\, y_{train}$. The encoded test point is the query, the encoded training points are the keys, and the training outcomes are the values. On the training data the encoded predictors are orthonormal, $F'_{train}F_{train} = I$. This does not hold out of sample.

Keeping only the leading eigenvectors gives principal component regression, and ridge regression changes the encoding. Replacing the linear weights with a softmax gives nonlinear attention, which the paper calls Attention Regression. It uses five heads, each with a learned matrix $\Omega_m = L_m L'_m$, and is compared with OLS, random forests, gradient boosting, and an MLP.

Reproducible Results on the Simulated Data

- **OLS as attention:** Compute the OLS predictions in the usual way and in the attention form. Show that they match. Check that $F'_{train}F_{train} = I$ and that $F'_{test}F_{test} \neq I$. For a few test points, plot the weights over the training points. Show that some weights are negative and that they sum to one when the model has an intercept.
- **Beyond OLS:** Keep only the leading L eigenvectors, for $L = 1, \ldots, 5$. Show that the predictions equal principal component regression. Then replace Λ with $\Lambda + \lambda I$. Show that the predictions equal ridge regression. Plot the test R^2 against L and against λ.
- **Attention Regression:** Build Attention Regression with 5 heads and $\lambda = 10^{-3}$. Train it with L-BFGS and early stopping. Initialize each L_m from the Cholesky factor of $(X'X + \lambda I)^{-1}$. Compare it with OLS, random forest, gradient boosting, and an MLP on all six data sets. Report the out-of-sample R^2.